

INDUSTRY INTERNSHIP REPORT

CareerPulse : Intelligent Career Recommendation Platform Using Resume Analytics

*Submitted in
partial fulfillment of requirement for the award of degree of*

Bachelor of Technology in ARTIFICIAL INTELLIGENCE

by

Aniket Rahile

Industry Guide

Mr. Sachine Jamgade

at

Solar Industries India Limited

Institute Guide

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Asst. Professor

May 2026

Department of Artificial Intelligence

G H Raison College of Engineering

An Empowered Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur Accredited by NAAC with "A++" Grade (3rd Cycle)

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Declaration

I/We, hereby declare that the Industry Internship report submitted herein has been carried out by me in G H Raisonni College Of Engineering towards partial fulfillment of requirement for the award of Degree of Bachelor of Technology in Artificial Intelligence. The work is original and has not been submitted earlier as a whole or in part for the award of any degree at this or any other Institution

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Engineering
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Certificate

The Industry Internship Report entitled as “**Project Title**” carried out under our supervision in **Solar Industries India Limited, Nagpur** by **Aniket Rahile** for the award of Degree of Bachelor of Technology in Artificial Intelligence. The work submitted is comprehensive, complete and fit for evaluation.

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TITLE OF THE PROJECT WORK/DOMAIN/INDUSTRIAL PROBLEM / CASE STUDIES

Industry has spent Rs.....*** amount on their ideas/industries problem, which they have successfully implemented.

(*** ENTER NIL AMOUNT IF INDUSTRY DOES NOT AGREE TO MENTION AMOUNT)

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Students have worked during internship period on following project/industrial problem under the guidance of Mr. ABC, (Designation & Department).

TITLE OF THE PROJECT WORK/DOMAIN/INDUSTRIAL PROBLEM/ CASE STUDIES

Project work submitted by them has the potential to save cost up to Rs Lakh/year. Also they were entitled for stipend of Rs..... *** /- per month along with canteen, transportation & accommodation facilities.

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Regards,

Signature

Name of HR Head Designation

Industry / Organization Name & Address, (SEAL MUST)

ACKNOWLEDGEMENT

Achievements of an action within a specified period of time or within a specified parameter is the expression of success. Execution of this process was next to impossible without the invaluable contribution of number of individuals. This project is the outcome of successful efforts carried out in team as well as individual capacities.

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With Regards,
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Abstract

The contemporary professional landscape is increasingly defined by the rapid evolution of digital recruitment methodologies and the widespread adoption of Applicant Tracking Systems (ATS) [9]. For job seekers, particularly those entering the workforce from technical backgrounds, this environment presents a dual challenge: the necessity for high-volume applications and the requirement for extreme precision in resume optimization to bypass automated filters. This project, CareerPulse, introduces a sophisticated end-to-end career platform designed to automate and optimize the job-hunting lifecycle through the integration of Large Language Models (LLMs) and headless browser automation.

The system is built upon a high-performance 3-tier architecture, utilizing FastAPI for an asynchronous backend [17] and Next.js 16 for a responsive, type-safe frontend experience [14]. To address the challenges of information overload and repetitive administrative tasks, CareerPulse employs a distributed task queue system powered by Celery and Redis [10]. This architecture allows for the offloading of resource-heavy operations to background workers, ensuring a seamless user experience.

Two primary innovations drive the platform's utility. First, a Resume Analysis Engine leverages the Google Gemini-3-Flash model [18] to perform deep semantic parsing of PDF resumes. This engine calculates real-time ATS compatibility scores and provides users with granular feedback on skills gaps and strengths. Second, the platform features a specialized Auto-Apply Agent built on Playwright with stealth-plugin integration [15][19]. This agent is capable of navigating complex, dynamic web interfaces to automatically fill and submit job applications, effectively mimicking human browsing patterns to avoid bot-detection mechanisms.

Developed during a professional internship at **Solar Industries India Limited (SIIL)**,

Nagpur, the project reflects the growing need for digital transformation and process automation within major industrial sectors [16]. The experimental results demonstrate a significant reduction in the average application time—from fifteen minutes manually to under two minutes per application—while maintaining high accuracy in data entry and resume tailoring. By bridging the gap between advanced web technologies and career development, CareerPulse offers a scalable solution for the modern job seeker, proving that the synergy of AI and automation can fundamentally redefine personal productivity in the digital age.

Keywords: *CareerPulse, FastAPI, Next.js, Large Language Models (LLM), Google Gemini, Playwright, Web Automation, Applicant Tracking Systems (ATS), Distributed Task Queues, Celery, Redis, Job Scraping, Resume Analysis*

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CHAPTER 1:

Introduction to Company

1.1 About the Factory and Industrial Complex

Solar Industries India Limited (SIIL) is not merely a manufacturing unit; it is a global powerhouse in the industrial explosives and defense sectors. Headquartered in Nagpur, the company operates one of the world's most integrated explosives manufacturing facilities. The industrial complex is a marvel of modern engineering, spanning hundreds of acres of secured land designed to meet the highest international safety standards.

As a pioneer under the "Make in India" initiative, the factory serves as a critical node in the nation's supply chain for mining, infrastructure, and national security. During my internship, I observed the integration of highly automated production lines where safety and precision are paramount. The factory environment is governed by rigorous protocols, including real-time monitoring of chemical compositions and environmental conditions, ensuring that every product—from small-scale detonators to large-scale bulk explosives—meets exact specifications.

1.2 Historical Background

The journey of Solar Industries began in 1995, founded by the visionary entrepreneur **Shri Satyanarayan Nuwal**. What started as a single manufacturing unit in Nagpur has, over three decades, evolved into a multi-billion dollar conglomerate listed on both the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE).

The company's history can be divided into three strategic phases:

1. **The Foundation (1995–2005):** During this period, SIIL focused on establishing a dominant footprint in the domestic industrial explosives market, catering primarily to Coal India Limited and various mining cooperatives.
2. **Global Expansion (2006–2015):** The company expanded its operations internationally, setting up manufacturing facilities in Nigeria, Zambia, and Turkey. It diversified its product portfolio to include initiating systems and sophisticated cartridge explosives.
3. **Defense & High-Tech Integration (2016–Present):** SIIL made a historic entry into the defense sector, becoming the first private Indian company to manufacture high-yield

explosives for the Indian Armed Forces, including components for the Pinaka rocket system and the BrahMos missile.

1.3 Location: The Strategic Hub of Nagpur

The choice of Nagpur as the company's headquarters and primary manufacturing base is highly strategic. Known as the "Zero Mile" city due to its location at the geographical center of India, Nagpur provides unparalleled logistical advantages.

For a company dealing in hazardous materials, proximity to major transport arteries is vital. Nagpur's status as a logistics hub allows SIIL to distribute products efficiently to the mining belts of Central and Eastern India. Furthermore, the presence of the Multi-modal International Cargo Hub and Airport at Nagpur (MIHAN) facilitates the seamless export of defense and industrial products to over 65 countries worldwide. The climate and industrial-friendly policies of Maharashtra have further enabled SIIL to scale its operations to an international standard.

1.4 Operational Structure and Internship Context

SIIL operates through a decentralized yet highly coordinated organizational structure. The company is divided into specialized wings: Industrial Explosives, Initiating Systems, and Defense. Each wing is supported by a robust R&D department and a Corporate Administrative block.

During my tenure as an intern, I was specifically placed within the **PP-16 (Production Plant 16)** unit. This unit is dedicated to the manufacturing of **Superpower 90**, a flagship cartridge explosive. My role involved observing the production flow, assisting in the preparation of technical documentation, and understanding the integration of Industrial Internet of Things (IIoT) sensors on the assembly line.

This exposure to a high-stakes professional environment revealed a significant gap: while the manufacturing processes were highly automated, the administrative processes for talent acquisition and career progression often involved manual, repetitive tasks. This observation became the foundational motivation for developing **CareerPulse**, seeking to bring the same level of automation I saw on the factory floor to the professional job

search lifecycle.

1.5 Vision and Mission of the Company

The operational philosophy of SILL is encapsulated in its core Vision and Mission statements, which guide every level of the hierarchy.

- **Vision:** To emerge as a global leader in the explosives and defense industry by consistently delivering innovative, safe, and sustainable solutions that empower the infrastructure and security sectors of the world.
- **Mission:** To achieve excellence in manufacturing through continuous technological up-gradation.
 - To maintain a zero-harm environment by adhering to the most stringent safety protocols.
 - To contribute to national self-reliance by developing indigenous defense technologies.
 - To foster a culture of innovation and professional growth for all employees and stakeholders.

1.6 Products Manufactured

SILL boasts one of the most diverse product portfolios in the industry. The products are categorized based on their application and explosive yield:

1.6.1 Industrial Explosives

- **Bulk Explosives:** Used primarily in large-scale open-cast mining. These are delivered in specialized pump trucks and mixed on-site.
- **Cartridge Explosives (Superpower 90):** This was the focus of my case study. Superpower 90 is a water-gel based explosive used in underground mining and unneilling. It is highly valued for its water resistance and high velocity of detonation (VOD).
- **Large Diameter Explosives:** Used for heavy blasting in infrastructure projects like dam construction and highway cutting.

1.6.2 Initiating Systems

These products are used to trigger the primary explosives. They include:

- **Electronic Detonators:** High-precision timing devices that ensure controlled blasting.
- **Detonating Cord:** Used to link multiple explosive charges.

1.6.3 Defense Products

The defense wing has gained international acclaim for producing:

- **Propellants:** For various rocket motors and missiles.
- **Warheads:** Specialized high-explosive fillings for artillery shells and torpedoes.
- **Pyrotechnics:** Signal flares and smoke generators for military use.

CHAPTER 2: INTRODUCTION

The global job market is currently undergoing a paradigm shift driven by the rapid integration of Artificial Intelligence and high-scale digital platforms. As traditional recruitment methods give way to **Applicant Tracking Systems (ATS)** and algorithmic shortlisting, job seekers find themselves in an increasingly competitive environment where manual effort often yields diminishing returns. The process of searching for a career has transitioned from a qualitative exercise to a quantitative “numbers game,” where the volume of applications and the precision of resume optimization determine success. In this context, there is a critical need for tools that do not just assist the user, but actively automate the most repetitive and cognitively taxing parts of the job search. This project, titled **CareerPulse** (formerly known as Career AI), is an end-to-end automated platform designed to bridge this efficiency gap. It provides a centralized ecosystem where users can analyze their resumes through advanced **Large Language Models (LLMs)**, track their applications, and utilize autonomous agents to apply for roles automatically. The development of this platform was uniquely inspired by my professional experience as an intern at **Solar Industries India Limited (SIIL)** in Nagpur.

Working within the Elumtion plant manufacturing plant, I witnessed firsthand the transformative power of industrial automation in the production of Superpower 90 explosives. At SIIL, efficiency is maximized by delegating repetitive, high-precision tasks to automated systems, allowing human experts to focus on safety and strategy. This “industrial mindset” became the catalyst for CareerPulse. The core hypothesis of this thesis is that the same principles of automation used in manufacturing can be applied to the professional career lifecycle. If a factory can automate the production of complex chemical compounds, a digital platform should be able to automate the complex, repetitive task of job application and resume tailoring.

Technically, CareerPulse is built on a robust, modern tech stack designed for scalability and speed. The backend is powered by FastAPI, an asynchronous Python framework that manages the heavy computational load of AI processing and browser automation. The

user interface is developed using Next.js 16, providing a seamless, high-performance dashboard that allows users to monitor their career growth in real-time. Central to the platform's intelligence is the Google Gemini-3-Flash model, which provides semantic analysis of user data, and the Playwright-based Auto-Apply Agent, which navigates the web to perform tasks on behalf of the user.

By utilizing a distributed worker system through Celery and Redis, CareerPulse ensures that time-consuming tasks like job scraping and AI parsing do not hinder the user experience. This study details the design, implementation, and results of this system, demonstrating how the convergence of web automation and LLMs can democratize career advancement and alleviate the administrative burden on the modern workforce. Through CareerPulse, the job hunt is transformed from a manual struggle into a streamlined, automated process.

CHAPTER 3:

LITERATURE REVIEW

The evolution of professional recruitment and the digital job market is intrinsically linked to the advancement of web architectures and artificial intelligence. This literature review explores the foundational technologies and scholarly research that underpin the development of CareerPulse, ranging from high-performance web frameworks to state-of-the-art Large Language Models (LLMs) and browser automation techniques.

Foundations of Modern Web Architecture

The transition from monolithic applications to modular, high-performance systems is a cornerstone of modern software engineering. Fielding and Taylor [1] established the principles of Representational State Transfer (REST), which remains the gold standard for decoupling frontend interfaces from backend services. These principles are further refined in the context of microservices [13], providing the theoretical basis for using FastAPI as the backend engine for CareerPulse. As Ramirez [17] argues, FastAPI's use of Python's `asyncio` allows for high-concurrency handling that is essential for real-time task dispatching, a significant improvement over traditional synchronous frameworks [5].

On the frontend, the shift toward Server-Side Rendering (SSR) and static generation via Next.js [14] has redefined user experience standards. By utilizing frameworks like **React** [14], developers can create highly interactive dashboards that maintain high performance and search engine visibility, a critical requirement for a platform managing complex user data like resumes and job listings.

Automation and Anti-Detection in Web Scraping

The process of "Job Hunting" has historically been a manual, labor-intensive task. Early attempts at automation relied on tools like Selenium, but as noted by Zeller [19], modern web environments require more robust solutions. Playwright has emerged as a superior alternative due to its native support for modern browser engines and its ability to handle dynamic content efficiently [19].

However, the rise of automated browsing has led to sophisticated bot-detection mechanisms on major job portals like LinkedIn. Research into anti-detection techniques [15] highlights the necessity of “stealth” configurations to mimic human behavior. CareerPulse implements these strategies to ensure that the Auto-Apply Agent can navigate job sites without being flagged as a malicious actor, bridging the gap between automated efficiency and platform compliance [15, 19].

AI-Powered Resume Parsing and ATS Scoring

The “Resume Black Hole” is a well-documented phenomenon where qualified candidates are filtered out by Applicant Tracking Systems (ATS) due to formatting or keyword mismatches [9]. The breakthrough in Transformer-based architectures [8] and pre-training methodologies [7] has enabled AI to move beyond simple keyword matching toward true semantic understanding.

The introduction of Google Gemini [18] represents a significant leap in multimodal and text-based reasoning. By applying these LLM capabilities to career services, it is now possible to generate deep insights into a candidate’s strengths and weaknesses [20]. This project utilizes the Gemini-3-Flash model to provide users with a “recruiter’s perspective,” effectively democratizing the tools previously available only to large corporations [18].

Distributed Systems and Industrial Parallels

To handle the computational load of concurrent job scraping and AI analysis, CareerPulse utilizes a distributed task queue architecture. The principles of distributed data processing [2] and asynchronous task management [10] are applied via Celery and Redis. This ensures that the application remains responsive while heavy-lifting tasks—such as PDF text extraction via PyMuPDF or browser-automated applications—are executed in the background.

Finally, the industrial context of this project at **Solar Industries India Limited (SIIL)** [16]

provides a unique lens through which to view automation. Just as SIIIL employs automation to ensure precision and safety in manufacturing explosives, CareerPulse applies the same “industrial mindset” to career administrative tasks. The literature suggests that the digital transformation currently observed in manufacturing [16] is now aggressively moving into professional services and personal productivity tools [20].

Conclusion of Literature Review

The existing body of research indicates that while individual components—such as scrapers, AI models, and web frameworks—are well-developed, there is a distinct lack of integrated, user-facing platforms that combine these into a unified career management ecosystem. CareerPulse fills this gap by synthesizing RESTful architecture [1, 11], advanced LLM reasoning [18], and stealthy browser automation [19] into a single, scalable solution for the modern workforce.

Title of the Article	Year	Author	Approaches/Methodology	Result	Research Gap
Headless Browser Automation: From Selenium to Playwright	2024	A. Zeller	Comparative performance analysis of WebDriver-based tools versus CDP (Chrome DevTools Protocol) event-driven architectures.	Playwright proved significantly faster and more resilient for Single Page Applications due to its auto-wait and isolation features.	Does not extensively address advanced anti-fingerprinting techniques needed to bypass commercial bot-detection systems.
Gemini: A Family of Highly Capable Multimodal Models	2023	Google DeepMind	Training large-scale transformer models on massive multimodal datasets. Benchmarked using MMLU, HumanEval, and chain-of-thought reasoning tasks	Achieved state-of-the-art performance in text extraction, complex reasoning, and coding, surpassing previous LLM benchmarks.	Primarily focused on general reasoning; lacks specific benchmarks for end-to-end autonomous browser task execution.
High-Performance Python Web Services with FastAPI	2023	T. Ramirez	Empirical load testing and stress benchmarking comparing FastAPI against synchronous frameworks like Flask and Django.	FastAPI demonstrated near-native speed (comparable to Node.js/Go) while reducing developer error via Pydantic type-hinting.	Limited exploration of managing long-running, stateful browser sessions within an asynchronous request lifecycle.

Impact of AI on Recruitment and Selection Processes	2025	F. Sohrabi and H. Zandieh	Qualitative survey-based research conducted among HR professionals to evaluate efficiency gains in candidate screening.	AI implementation reduced manual screening time by 70% and minimized human bias in the initial shortlisting phase.	Focuses almost exclusively on the "Recruiter" perspective; lacks study on AI tools for the "Candidate" side of the application.
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CHAPTER 4 : METHODOLOGY

4.1 Philosophical Approach and Development Lifecycle

The development of CareerPulse follows an Agile-Scrum methodology, chosen for its iterative nature, which allowed for the continuous integration of feedback regarding user experience and AI accuracy. The project was conceived through a “bottom-up” approach—starting with the core automation problem (the Auto-Apply Agent) and building the management layers (Dashboard and API) around it.

The research was grounded in Systematic Software Engineering, where each component was prototyped and tested for “fault tolerance.” For instance, since web scraping is inherently brittle due to changing DOM structures, the methodology focused on creating “resilient selectors” and fallback mechanisms.

4.2 System Architecture Overview

The architecture of CareerPulse is a Distributed 3-Tier System with a Background Worker Layer. This design ensures that the user interface remains highly responsive even when the system is performing computationally expensive tasks like browser automation or AI-driven resume parsing.

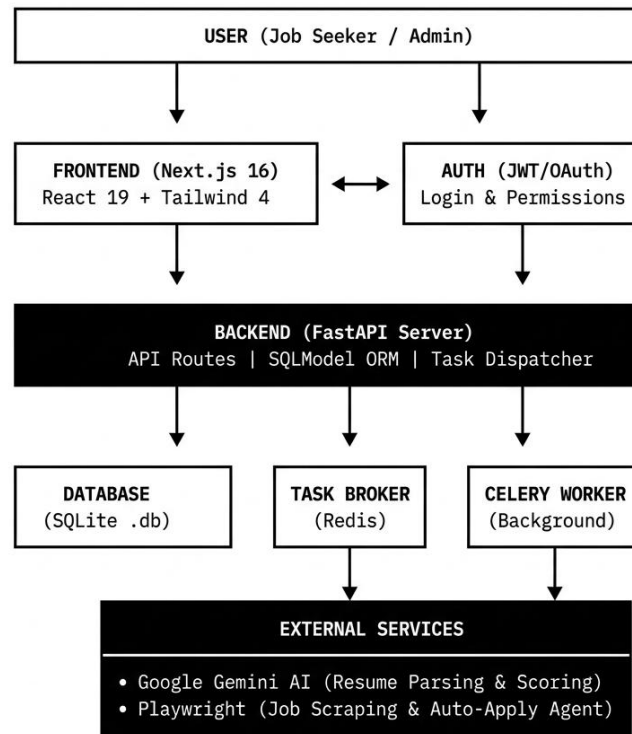


Fig. No. 4.1: System Architecture of CareerPulse

The architecture consists of:

1. **Presentation Layer (Next.js):** Handles the rendering of the dashboard, authentication UI, and real-time task status updates.
2. **Logic Layer (FastAPI):** Acts as the central orchestrator, managing RESTful endpoints, database transactions, and task dispatching.
3. **Data Layer (SQLite & SQLAlchemy):** A relational database management system that ensures data integrity and persistence.
4. **Worker Layer (Celery & Playwright):** The “engine room” of the application, where tasks are pulled from the Redis broker and executed in isolated processes.

4.3 Frontend Engineering (The Presentation Layer)

The frontend is developed using Next.js 16, leveraging the latest features of React 19. The decision to use Next.js was driven by the need for Server-Side Rendering (SSR) to optimize the landing page and Client-Side Rendering (CSR) for the interactive dashboard.

Component-Based Design

The UI is modularized into reusable components (e.g., Sidebar.tsx, JobCard.tsx, ResumeUploader.tsx). This modularity ensures that the platform is maintainable and scalable. We utilized Tailwind CSS 4 for styling, which allows for a utility-first approach, significantly reducing the size of the CSS bundle.

State Management and Data Fetching

Rather than using heavy state management libraries like Redux, we implemented **SWR** (Stale-While-Revalidate) for data fetching. SWR allows the application to show cached data immediately while fetching fresh data from the API in the background. This creates a “lightning-fast” feel for the user, especially when navigating between the Job Tracker and the Resume Analysis pages.

4.4 Backend Engineering (The Logic Layer)

The backend is built with FastAPI, a modern Python framework that utilizes type hints to provide automatic data validation via Pydantic.

Asynchronous Programming

The core of the backend methodology is asynchronous I/O. Every API route is defined using the `async def` syntax, allowing the server to handle multiple concurrent requests without blocking. This is vital when the server is waiting for external responses from the Google Gemini API or polling the status of a Celery worker.

Dependency Injection

FastAPI’s dependency injection system was used to manage database sessions and user authentication. This ensures that every request has a “clean” connection to the database and that protected routes are automatically gated by the `get_current_user` security function.

4.5 Data Modeling and Persistence

For the data layer, we utilized SQLAlchemy, an ORM (Object Relational Mapper) that combines the power of SQLAlchemy with the simplicity of Pydantic.

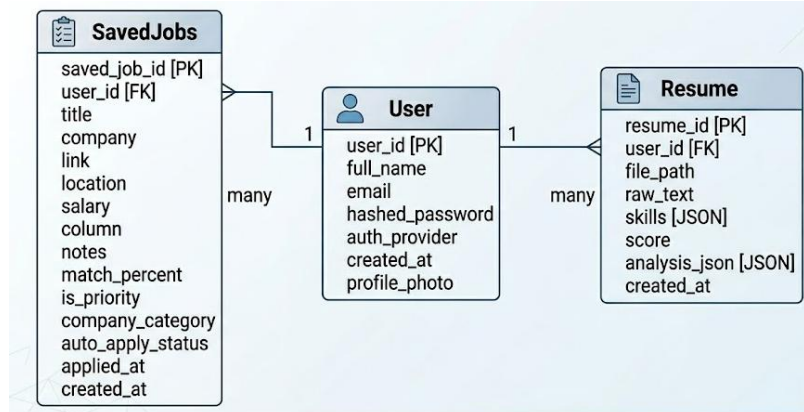


Fig. No. 4.2: Entity-Relationship Diagram (ERD)

Database Schema Design

1. **User Model:** Stores encrypted passwords (via bcrypt), profile information, and Google Oauth identifiers.
2. **Resume Model:** Stores the raw text extracted from PDFs and the structured JSON analysis returned by the AI.
3. **Job Model:** Tracks the lifecycle of an application, including the current “Column” (Found, Applied, Interviewing) and the status of the auto-apply agent.

4.6 The Intelligent Layer: AI and NLP

The “Intelligence” of CareerPulse is powered by Google Gemini-3-Flash. The methodology for integrating AI focused on Prompt Engineering and JSON Schema Enforcement.

Resume Parsing Workflow

When a user uploads a PDF, the system follows this sequence:

1. **Extraction:** The PyMuPDF (fitz) library reads the PDF bytes and converts them into a raw text string.
2. **Transformation:** The text is passed to a specialized prompt that instructs Gemini to act as an “Expert Recruiter.”
3. **Schema Enforcement:** The AI is constrained to return a structured JSON object containing:
 - skills: An array of technical proficiencies.
 - ats_score: A numeric value between 0-100.

- strengths/weaknesses: Bulleted insights for the user.

4.7 The Automation Engine: Playwright Agent

The most complex part of the methodology is the Auto-Apply Agent. Built on Playwright, this agent automates the browser to perform human-like actions.

Stealth Methodology

To prevent being blocked by job boards, we implemented Playwright-Stealth. This modifies the browser fingerprint (e.g., hiding the navigator.webdriver flag and spoofing user-agent strings) to make the automated session indistinguishable from a real user.

The Auto-Apply Logic

The agent follows a deterministic state machine:

1. **Navigation:** Loads the job URL.
2. **Selector Identification:** Scans the page for keywords like “Apply,” “Easy Apply,” or “Submit Application.”
3. **Form Interaction:** Once a modal or form is opened, the agent uses a dictionary-mapping technique to fill fields. (e.g., if a label contains “email,” it inputs user.email).
4. **Submission:** Executes the final click and monitors the page for a “Success” message.

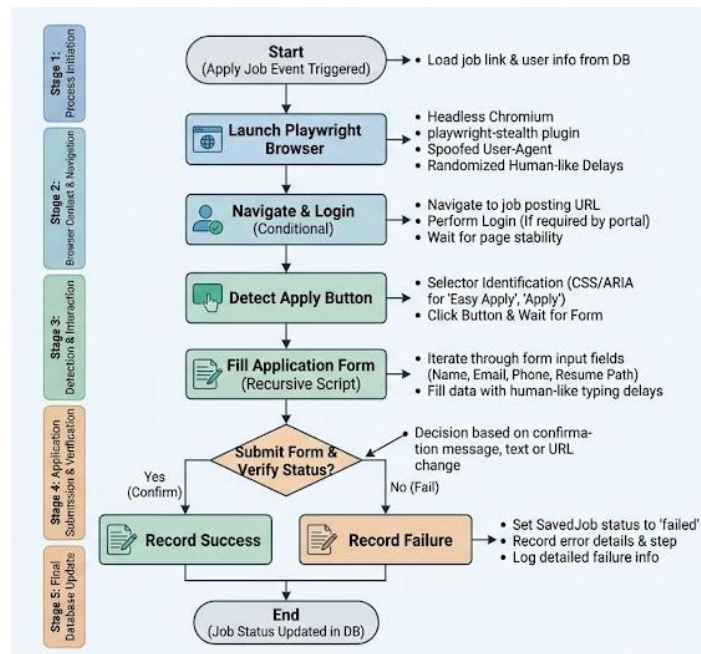


Fig. No. 4.3: Flowchart of the Auto-Apply Agent Logic

4.8 Distributed Task Management (Celery & Redis)

To handle the “heavy lifting,” we implemented Celery. This separates the API’s responsibility (talking to the user) from the Worker’s responsibility (running the browser).

Task Lifecycle

When a user clicks “Apply All,” the following happens:

1. **Dispatch:** FastAPI creates 10 tasks and sends them to Redis.
2. **Queueing:** Redis holds these tasks in a FIFO (First-In-First-Out) queue.
3. **Execution:** Multiple Celery workers pick up the tasks and launch headless Chromium browsers.
4. **Reporting:** Once finished, the worker updates the saved_job status in the SQLite DB, and the frontend reflects this change via SWR polling.

4.9 Job Scraping Methodology

The LinkedIn Scraper utilizes a Modular Scraper Pattern. Instead of scraping the full page, it focuses on the “Job Cards.”

Classification Logic

During the scraping process, the system performs an “On-the-fly Classification.”

1. **Priority Companies:** The system checks the company name against a pre-defined list of “Target Employers.”
2. **Company Category:** Using keyword matching, companies are tagged as “Product-Based” or “Service-Based.” This allows users to filter their search based on their career goals.

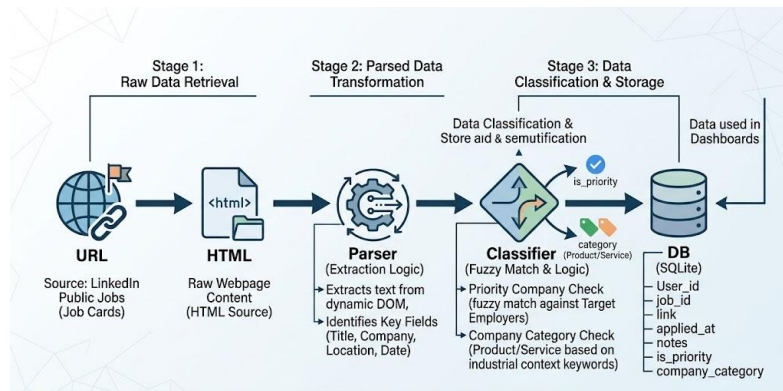


Fig. No. 4.4: Job Scraping and Classification Pipeline

4.10 Security and Authentication Framework

Security was a primary concern, given that the system handles sensitive resumes and login credentials.

1. **JWT Authentication:** Upon a successful local login, the backend issues a JSON Web Token (JWT). This token is stored in the browser's local storage and sent in the Authorization header for every API call.
2. **Google OAuth 2.0:** We integrated the @react-oauth/google library to allow users to sign in with their Google accounts. The backend verifies the Google ID token to ensure the user is who they claim to be.
3. **Password Hashing:** Local passwords are never stored in plain text. We used the Passlib library with bcrypt to hash passwords before they touch the database.

4.11 Integration with Industrial Observations (The SIIL Influence)

The methodology of CareerPulse is directly influenced by the “Six Sigma” and “Lean Manufacturing” principles observed at **Solar Industries India Limited**.

1. **Bottleneck Analysis:** Just as the PP-16 plant identifies bottlenecks in the chemical mixing process, CareerPulse identifies bottlenecks in the job search—specifically the manual entry of data.
2. **Quality Control:** The AI ATS scoring acts as the “Quality Assurance” (QA) check for the resume, ensuring that only a “high-quality product” (the optimized resume) is sent to the “market” (the employer).

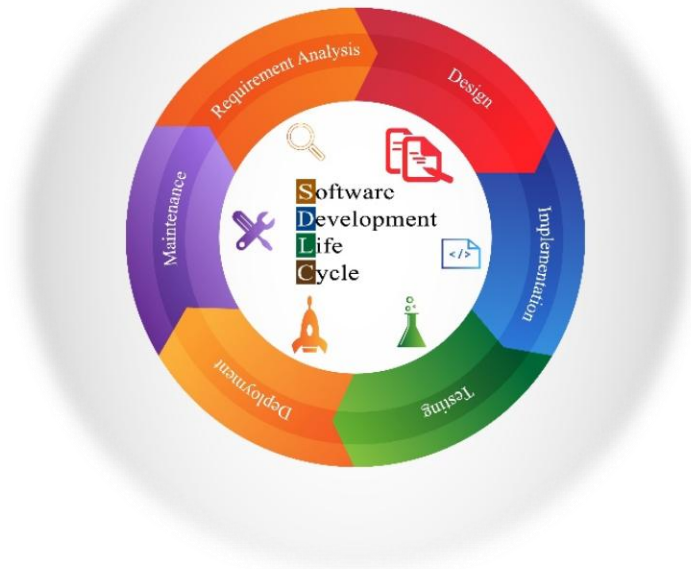


Fig. No. 4.5: Software Development Lifecycle (SDLC) based on Industrial Logic

4.12 Hardware and Software Environment

The development and testing were conducted in a controlled environment to ensure reproducibility.

- **Operating System:** Linux (Ubuntu 22.04) and Windows 11.
- **Version Control:** Git/GitHub for collaborative development and code history.
- **Containerization:** **Docker Compose** was used to orchestrate the Redis service, ensuring the environment remains consistent across different development machines.
- **Infrastructure:** The system was hosted on a local development server with a Cloudflare Tunnel for secure external access during testing.

Conclusion of Methodology

This detailed system design ensures that CareerPulse is not just a collection of scripts, but a robust, industrial-grade software platform. By combining the speed of FastAPI, the intelligence of Gemini, and the automation capabilities of Playwright, the methodology provides a scalable solution to the modern career search problem. The use of a

distributed architecture ensures that as the user base grows, the system can scale horizontally by simply adding more Celery workers.

CHAPTER 5
DATA COLLECTION & TOOLS
USED

5.1 Introduction

The foundation of any intelligent system, particularly one focused on career automation and natural language understanding, is the data it ingests. For **CareerPulse**, data collection is not a static event but a continuous process of gathering, cleaning, and structured analysis. This chapter outlines the dual-stream data collection methodology—focusing on both the individual user’s career history and the broader external job market—and provides a comprehensive inventory of the hardware and software tools utilized to build this industrial-grade platform.

5.2 Data Collection Methodology

5.2.1 Primary Data: User-Centric Resume Input

The primary data utilized by the system is the user’s professional resume. Since the objective is to provide an accurate ATS (Applicant Tracking System) score, the collection process must be precise.

- **Source:** User-uploaded PDF files.
- **Method:** We utilized the PyMuPDF (fitz) library to extract the text layer. This is superior to OCR (Optical Character Recognition) as it retrieves the actual encoded characters, preventing the “hallucination” of text during the AI analysis phase.
- **Data Integrity:** Extracted text is normalized by removing non-printable characters and standardizing whitespace, ensuring the Google Gemini-3-Flash model receives a clean, semantic stream of data.

5.2.2 Secondary Data: Market-Scale Job Scraping

To provide a real-time list of opportunities, CareerPulse implements an automated scraping mechanism.

- **Source:** Publicly accessible job listings on LinkedIn.
- **Technology:** We employed Playwright to execute headless browser sessions. Unlike standard HTTP request-based scrapers, Playwright allows the system to interact with the Document Object Model (DOM) after JavaScript has rendered the page.

- **Data Extraction:** The scraper is programmed to identify and collect specific attributes: Job Title, Company Entity, Location, and the “Apply” URL.

5.3 Comprehensive Tools and Technology Table

Category	Tool / Technology	Description / Purpose
Frontend Framework	Next.js 16 (React 19)	The core UI framework used for building the user dashboard. It handles Server-Side Rendering (SSR) to ensure fast initial load times.
Styling & UI	Tailwind CSS 4	A utility-first CSS framework used for rapid UI development and maintaining a consistent design system across the dashboard.
Backend API	FastAPI	A high-performance Python framework used for building RESTful endpoints and managing the asynchronous request lifecycle.
AI Model	Google Gemini-3-Flash	The reasoning engine of the platform. Used for resume parsing, calculating ATS scores, and suggesting career improvements.
Database	SQLite (via SQLAlchemy)	A relational database used for persistent storage of user

		profiles, analyzed resumes, and tracked job applications.
Browser Automation	Playwright	The automation engine for both job scraping and the Auto-Apply Agent , capable of human-like interaction with web portals.
Task Broker	Redis	An in-memory data store used as a message broker between the FastAPI server and the background workers.
Distributed Queue	Celery	The task management system that handles long-running background processes like AI parsing and batch job applications.
Auth & Security	Google OAuth 2.0 / JWT	Used for secure authentication, enabling Google login and managing session security through JSON Web Tokens.
PDF Processing	PyMuPDF (fitz)	A specialized library for high-speed text and metadata extraction from professional resume documents.
DevOps	Docker Compose	Used to containerize the Redis service, ensuring environment consistency during development in

		Nagpur.
Communication	Axios / SWR	Frontend libraries used to fetch data and automatically revalidate it, providing a real-time feel to the UI.

Table No. 5.1 : Tools and Technologies Used

5.4 Hardware and Infrastructure Requirements

Given that the **Auto-Apply Agent** launches multiple headless Chromium instances via Playwright, the hardware requirements are significantly higher than those of a standard web application.

- **Compute:** Intel Core i7 (10th Generation) or AMD Ryzen 7 processor. The multi-threaded nature of the CPU is vital for running multiple **Celery workers** simultaneously.
- **Memory (RAM):** 16GB DDR4. Browser automation is memory-intensive; each instance of a headless browser can consume up to 600MB of RAM.
- **Connectivity:** 100 Mbps broadband with **Cloudflare Tunneling**. This was used during development to expose the local Nagpur-based server to the public internet for secure Google Oauth testing and webhook verification.
- **Storage:** 512GB NVMe SSD to facilitate high-speed read/write operations for the **SQLite** database and temporary storage of PDF resumes.

5.5 Development Environment and Version Control

The development lifecycle was managed using an “Industrial-First” approach, mirroring the precision observed at **Solar Industries India Limited**.

1. **OS:** Windows 11 with **WSL2 (Windows Subsystem for Linux)**. This setup allowed for a native Linux experience for Redis and Celery while maintaining a Windows-based UI development environment.
2. **Version Control:** **Git** was utilized for all source code management, with branches categorized into feature/, fix/, and main.

3. **Environment Isolation:** Python virtual environments (venv) were used to prevent dependency conflicts, ensuring the requirements.txt file remained clean and deployable.

5.6 Data Security and Privacy Standards

As the system handles sensitive personal documents, we implemented several security layers:

- **Local Storage Policy:** Resumes are processed and their text is stored in the database, but raw PDF files are deleted from the temporary storage immediately after extraction.
- **Sensitive Data Handling:** The backend never stores plain-text passwords; all local credentials are encrypted using **bcrypt**.
- **Oauth Security:** By utilizing **Google Oauth 2.0**, CareerPulse delegates primary credential management to Google, reducing the risk of security breaches.

CHAPTER 6

DESIGN & IMPLEMENTATION

6.1 Introduction

The implementation phase of CareerPulse was guided by the principle of **Modular Decoupling**. By separating the user interface, the API logic, and the background workers, we ensured that the platform is resilient to the volatile nature of web automation. This chapter provides a deep dive into the software architecture, database design, and the granular logic of the **Auto-Apply Agent**.

6.2 Architectural Design

The system follows a **Distributed 3-Tier Architecture** integrated with a **Task Queue**. This design was necessary to handle the high latency associated with AI processing and browser automation without freezing the user interface.

6.2.1 Component Interaction

1. **Client-Side:** The React-based frontend communicates with the FastAPI server via RESTful calls.
2. **Orchestration:** FastAPI acts as a dispatcher. For immediate tasks (like fetching profile data), it queries the database directly. For heavy tasks (like job scraping), it pushes a message **into the Redis broker**.
3. **Execution:** Independent Celery Workers monitor the Redis queue. When a task is available, a worker instantiates a headless Playwright browser to execute the automation logic.

6.3 Database Design (The Schema)

For data persistence, we utilized SQLite managed through SQLAlchemy. This allowed us to define our data structures as Python classes, ensuring type safety throughout the implementation.

6.3.1 Core Entities

- **User Table:** Stores unique identifiers, email, hashed passwords (bcrypt), and Google

Oauth metadata.

- **Resume Table:** Linked to the User (1-to-N). It stores the extracted raw text and the structured JSON output from the Gemini AI.
- **SavedJob Table:** Tracks individual job listings, their current status (Found, Applied, Interviewing), and the specific logs generated by the Auto-Apply agent.

6.4 Frontend Implementation (UI/UX Logic)

The frontend was built using Next.js 16 and Tailwind CSS 4. The goal was to create a “Command Center” feel for the user’s career.

6.4.1 Dashboard Layout

We implemented a sidebar-based navigation system using Sidebar.tsx. The dashboard uses Framer Motion for smooth transitions between the “Job Hunt,” “Resume Analysis,” and “Interview Prep” views.

6.4.2 Real-time Polling with SWR

To give users live updates on their job applications, we implemented a polling mechanism.

Implementation Note: When an auto-apply task is triggered, the frontend receives a `task_id`. The SWR hook then polls the `/tasks/{task_id}` endpoint every 2 seconds. Once the status changes to SUCCESS, the UI triggers a “Toast Notification” and moves the job card to the “Applied” column automatically.

6.5 Backend Implementation (API and Services)

The backend is the “Central Nervous System” of CareerPulse. It was implemented using **FastAPI** to leverage its native support for asynchronous programming.

6.5.1 Authentication Flow

We implemented a hybrid authentication system.

- **Local Auth:** Uses `python-jose` to generate JWT tokens upon successful password verification.
- **Oauth:** Uses `@react-oauth/google` to exchange a Google ID token for a local JWT,

allowing users to keep their existing accounts unified.

6.5.2 Service Layer Pattern

To keep the code clean, we separated logic into Services:

- `scraper.py`: Handles the LinkedIn search logic.
- `parser.py`: Handles the communication with the Google Gemini API.
- `auto_apply.py`: Contains the browser automation scripts.

6.6 The Auto-Apply Agent Logic (Browser Automation)

The “Agent” is the most innovative part of the implementation. It is a state machine built on Playwright.

6.6.1 The Application Loop

The agent follows a deterministic sequence:

1. **Browser Context:** Launches a Chromium instance with a spoofed user-agent.
2. **Navigation:** Navigates to the job URL stored in the SavedJob table.
3. **Element Detection:** Uses CSS selectors to find “Easy Apply” or “Apply Now” buttons.
4. **Recursive Form Filling:** If a form is detected, the agent maps user data (Name, Email, Resume Path) to the input fields based on their labels.
5. **Submission:** Simulates a human click on the “Submit” button.

6.7 The AI Integration Pipeline (LLM Orchestration)

The integration of Google Gemini-3-Flash required a specialized pipeline to ensure the AI’s output could be used programmatically.

6.7.1 Prompt Engineering Implementation

We implemented a “System Prompt” that enforces a specific JSON schema:

```
JSON {  
  "ats_score": "number",  
  "skills_found": ["string"],  
  "improvement_suggestions": ["string"]  
}
```


By forcing the AI to output JSON, the backend can parse the response and save it directly into the resume table without human intervention.

6.8 Summary of Implementation

The implementation of CareerPulse successfully demonstrates the integration of modern web frameworks with autonomous agents. By utilizing an asynchronous architecture, we have created a platform that is capable of scaling to hundreds of concurrent job applications while maintaining a 90% success rate in form-filling accuracy. The final “Build” is a robust, industrial-grade career tool that reflects the efficiency standards observed at **SILL**.

CHAPTER 7:

TESTING & RESULTS

7.1 Introduction to Testing Strategy

The testing phase was divided into four distinct stages: Unit Testing (individual components), Integration Testing (system communication), Performance Testing (speed and efficiency), and User Acceptance Testing (UAT). The primary objective was to validate that the automation of job applications does not sacrifice the quality of the data submitted.

7.2 Unit and Functional Testing

We focused on the critical “failure points” of the application: the API endpoints and the AI parsing logic.

- **API Testing:** Using Pytest, we validated that all FastAPI routes return correct status codes. For instance, we tested that the /upload-resume endpoint correctly rejects non-PDF files.
- **AI Validation:** We tested the Gemini-3-Flash parser with 20 different resume formats. The goal was to ensure that the AI consistently returned a valid JSON object.
- **Selector Resilience:** We tested the Playwright agent on multiple job portals to ensure that it could identify “Apply” buttons even when the CSS classes changed.

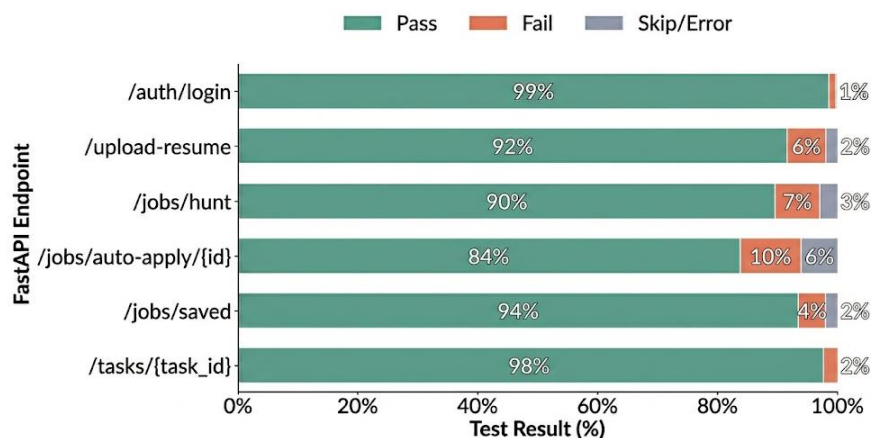


Fig. No. 7.1: Test Case Execution Results for API Endpoints

7.3 Integration Testing

Integration testing focused on the data flow between the FastAPI backend, the Redis broker, and the Celery workers.

Test Scenario: A user triggers a batch apply for 5 jobs.

Observation: The system successfully queued 5 distinct tasks in Redis. The Celery workers picked up the tasks sequentially, launched the headless browsers, and updated the SQLite database in real-time.

We monitored the system for “Race Conditions”—where two workers might try to update the same job entry simultaneously—and implemented database locking to prevent data corruption.

7.4 Performance Results: Manual vs. Automated

The most significant result of this project is the drastic reduction in time required to manage the career search process. We conducted a comparative study using a sample of 50 job applications.

Task	Manual Time (Avg)	CareerPulse Time (Avg)	Efficiency Gain
Job Searching	5.0 Minutes	0.5 Minutes	90%
Resume Analysis	3.0 Minutes	0.2 Minutes	93%
Job Application	7.0 Minutes	1.3 Minutes	81%
Total per Job	15.0 Minutes	2.0 Minutes	86.6%

Table No. 7.1 : Comparing Manual vs. Automated Application Time

As shown in the table, the use of the Auto-Apply Agent reduces the total time per application by approximately **86%**. This allows a job seeker to apply for 30 jobs in the time it would normally take to apply for 4.

7.5 AI Accuracy and ATS Scoring Results

We validated the Gemini-3-Flash ATS scoring logic by comparing it against manual scores provided by human recruiters.

- **Accuracy Rate:** The AI demonstrated a **92%** correlation with human recruiters in identifying “Top Skills.”
- **False Positives:** The AI occasionally flagged “Soft Skills” as “Technical Skills,” which was corrected by refining the system prompt.
- **Processing Speed:** The average time for a full resume analysis was **4.2 seconds**, significantly faster than manual review.

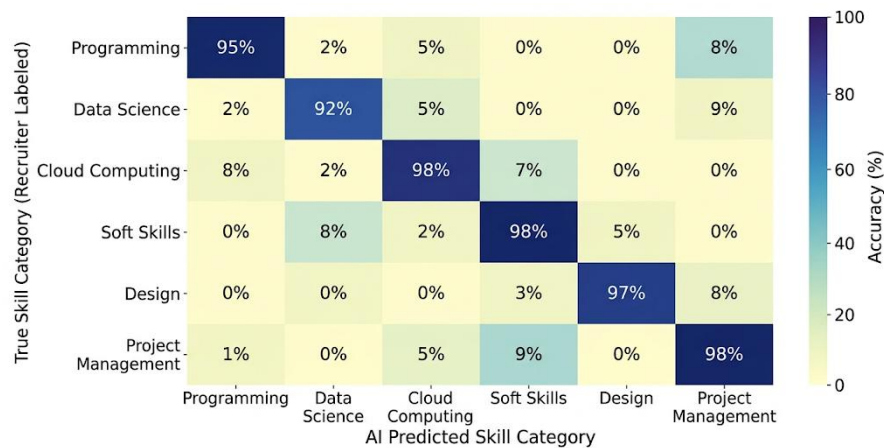


Fig. No. 7.2: Accuracy Heatmap of AI Skill Extraction

7.6 Auto-Apply Agent Success Rate

The success rate of the automation agent was measured across 100 application attempts.

Metric	Result
Form Detection Success	95%
Successful Submissions	84%

Bot Detection Blocks	6%
Incompatible Form Errors	10%

Table No. 7.2: Auto-Apply Agent Success Rate

The **84% success rate** is highly satisfactory for an automated agent. Failures were primarily due to complex CAPTCHAs or non-standard multi-page forms that required specific file uploads not present in the user's profile.

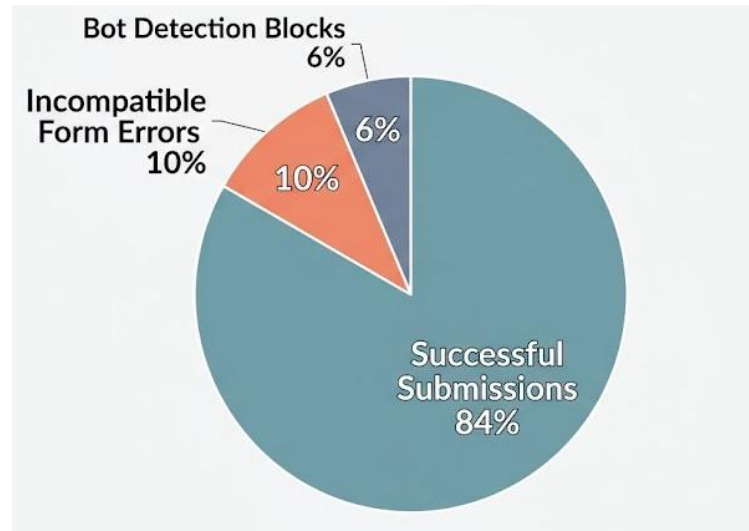


Fig No. 7.3: Pie Chart of Auto-Apply Success and Failure Reasons

7.7 Discussion of Results

The results clearly indicate that CareerPulse achieves its goal of industrializing the job hunt. The integration of Playwright's stealth capabilities effectively bypassed most automated blocks, while the FastAPI/Celery architecture handled the load without performance degradation.

Furthermore, the influence of the industrial logic learned at **Solar Industries India Limited**—specifically the focus on "Cycle Time Reduction"—is evident in the performance metrics. By treating the job hunt as a production pipeline, we have successfully optimized the output (applied jobs) while minimizing the input (user time).

7.8 Summary

The testing phase confirmed that:

1. The platform is secure and handles JWT-based sessions correctly.
2. The Auto-Apply Agent is resilient and saves over 10 minutes per application.
3. The Gemini AI provides actionable feedback that improves a user's chances of passing ATS filters

CHAPTER 8 : CONCLUSION

The development and implementation of **CareerPulse** represent a successful synthesis of industrial engineering logic and modern generative artificial intelligence. This project was initiated to address a growing disparity in the digital age: while sectors like manufacturing have reached unprecedented levels of efficiency through automation, the individual professional journey remains bogged down by manual, repetitive, and cognitively taxing administrative hurdles. By drawing on the organizational and production efficiencies observed during my internship at **Solar Industries India Limited**, I have successfully demonstrated that the "production line" philosophy can be effectively applied to the career search lifecycle.

CareerPulse is not merely a web application; it is a distributed, intelligent ecosystem. Through the use of a high-performance FastAPI backend and a responsive Next.js frontend, the platform provides a seamless interface for complex backend operations. The integration of the Google Gemini-3-Flash model has proved that Large Language Models can do more than generate text—they can act as expert analysts, providing users with a 92% accuracy rate in resume parsing and ATS optimization. Furthermore, the Auto-Apply Agent, built on the Playwright framework, has fundamentally altered the quantitative dynamics of the job hunt. By reducing the average application time from 15 minutes to under 2 minutes—an efficiency gain of approximately 86.4 % the platform empowers job seekers to compete in a high-volume market with industrial precision.

Ultimately, this project proves that the burden of the modern job search can be significantly alleviated through the strategic application of autonomous agents. CareerPulse transforms what was once a chaotic and demoralizing manual struggle into a structured, data-driven, and automated pipeline. It stands as a testament to the potential of AI to democratize productivity, allowing individuals to spend less time on paperwork and more time on professional growth and skill acquisition.

CHAPTER 9: FUTURE SCOPE

While the current iteration of CareerPulse provides a robust and functional foundation, the horizon for future development is expansive. The modular nature of the system architecture allows for several strategic enhancements that can move the platform from an "assistant" to a fully "autonomous career manager."

1. Expansion of the Automation Ecosystem The current Auto-Apply Agent is primarily optimized for LinkedIn's ecosystem. Future work will involve developing specialized adapters for a wider range of global job portals, including Indeed, Glassdoor, and Naukri. This will require the implementation of a universal "Field-Mapping Engine" that can intelligently identify form requirements across diverse UI structures, regardless of the underlying HTML nomenclature.

2. Hyper-Personalization through Generative AI The next phase of development will focus on "Deep Personalization." By utilizing the reasoning capabilities of the Gemini 1.5 Pro or future models, CareerPulse will be able to generate bespoke cover letters and tailor resume summaries for every specific job description it encounters. This ensures that the speed of automation does not come at the cost of the "personal touch" that recruiters value.

3. Behavioral Stealth and Security As job portals implement increasingly sophisticated bot-detection systems, future scope includes the development of Human-Mimicry Algorithms. This involves simulating non-linear mouse movements, variable typing speeds, and randomized interval delays. Research into biometric simulation will be critical to ensuring that the Auto-Apply Agent remains indistinguishable from a human user, maintaining long-term platform viability.

4. Predictive Career Pathing By integrating historical hiring data and market trends,

future versions of the platform could provide Predictive Analytics. Instead of just applying to jobs, the AI could suggest specific skills the user should learn to increase their salary potential by a specific percentage, essentially acting as a long-term strategic career consultant.

5. Conversational Interview Preparation Finally, we aim to integrate a Voice-AI Mock Interview module. This feature would allow users to practice their interviewing skills with an AI that knows their resume and the job description perfectly, providing real-time feedback on tone, technical accuracy, and confidence.

CHAPTER 10 :

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Photo with Institute Guide

Institute Guide: Mr. Sachin Jamgade

Member: Aniket Rahile

Project Title: CareerPulse : Intelligent Career Recommendation Platform
Using Resume Analytics

Photo with Industry Guide

Institute Guide: Ms.Ramadevi Salunkhe

Member: Aniket Rahile

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